MILITARY SPECIFICATION

PROJECTILE FOR CARTRIDGE, 105-MILLIMETER, M360, GB-FILLED; WITHOUT FUZE AND BURSTER

1. SCOPE

1.1 This specification covers the projectile for cartridge M360 filled with GB (for the 105-mm howitzer) without explosive components.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

75-14-705 - Projectile, Gas, Nonpersistent, GB, For Cartridge, 105 MM, M360; Loading Assembly and Marking Diagram.
76-18-44 - Pallet for Projectile, Chemical, 105 MM, M360, Assembly, Details, Packing.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are a part of the Code of Federal Regulations (Revised 1956) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-90 (Rev. 1956)")

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain, prior to their use, that all materials and components have met their respective specifications.

3.2 Rust prevention of components. The rust preservative of the projectile for cartridge applied for protection for storage and transportation shall not be removed until just prior to filling.

3.3 Filling level. The height of GB in the filled projectile (see 6.2 and 6.3) shall be within the limits shown in table I when examined as specified in 4.6.3.

Table I. Filling level

Temperature range °F	Level of liquid below nose of projectile	
	minimum	maximum
40 to 50	5.48	6.06
50 to 60	5.45	6.03
60 to 70	5.42	6.00
70 to 80	5.39	5.96
80 to 90	5.36	5.93
90 to 100	5.33	5.89

Calibration factors:

10.0 percent void at 40°F in minimum projectile.
Expansion factor for GB = 0.00057 ml/°F/ml of agent.

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3.4 Closing. The projectile shall be closed as shown on Drawing 75-14-705. From the time of filling until completion of test specified in 4.6.2, the projectile shall be kept in a nose-up position (see 6.3).

3.5 Agent decontamination. The exposed surfaces of the projectile shall be free from GB contamination (see 4.6.1 and 6.4).

3.6 Helium leakage. The projectile shall not leak helium through the closure at a rate greater than 1×10^{-6} atmospheric cubic centimeter (cc) per second when tested as specified in 4.6.2.

3.7 Workmanship. The projectile shall be free from damage that could affect subsequent assembly and functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Joint responsibility. Tests prescribed by 4.6 will be performed jointly by the Government and the supplier.

4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.1.1 Filling level. When an alternative procedure is approved for use in determining the filling level, the procedure shall be monitored in accordance with 4.6.3 to insure that the requirements of this specification are satisfied.

4.2.2 Supplemental evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

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4.3 Lotting. A lot shall consist of the projectiles filled and assembled by one manufacturer, at one plant, using the same materials, under essentially the same filling conditions. However, no more than one lot of agent, one lot of metal parts as identified by one manufacturer, and one metal part interfix number shall be represented in any one lot of filled projectiles. For ballistic considerations a sublot shall consist of the projectiles having the same zone designation (see 5.1).

4.4 Sampling. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.5 Examination. Sample projectiles and the level A preparation for delivery thereof shall be examined in accordance with the classification of defects and Standard MIL-STD-105.

4.5.1 Classification of defects.

4.5.1.1 Projectile, gas, nonpersistent, GB, for cartridge, 105 mm, M360, loading assembly (Dwg 75-14-705).

Categories	Defects	Acceptance standards
*Critical:		
1	Contamination	4.6.1
2	Helium leakage	4.6.2
3	Filling excessive	4.6.3
Major:	AQL 1.0 percent defective	
101	Component missing	
102	Burster well seal depth not within tolerance specified	
103	Bourrelet incorrect (after painting)	
104	Marking incorrect	
105	Damage (rotating band if unprotected)	
Minor:	AQL 2.5 percent defective	
201	Filling insufficient	4.6.3
202	Protective finish missing or incorrect	
203	Marking missing or illegible	

*Each projectile in the lot shall be examined for these defects.

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4.5.1.2 Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Pallet incorrect
102	Quantity per pallet incorrect
103	Strapping loose or missing
104	Pallet broken or wracked
105	Marking missing, illegible, or incorrect

4.6 Tests.

4.6.1 GB decontamination. Contamination shall be determined by the use of an approved Chemical Corps GB agent detector system.

4.6.2 Helium leakage. Prior to closing, the void above the agent shall be evacuated to a pressure not greater than 7.5 pounds per square inch absolute (psia) and filled with helium to a pressure of 22 psia to insure a partial pressure due to helium of not less than 15 pounds per square inch gage (psig) after closing. The closure area shall be sealed off by a test head manifolded to a vacuum system and a mass spectrometer type helium leak detector. The test head shall be evacuated to and maintained at a pressure lower than 20 microns of mercury during the test. The leak detector shall be calibrated to respond to a leak of 1×10^{-6} atmospheric cc per second of helium at the test head when a standard leak is introduced into the system at that point.

4.6.3 Filling level. The height of the GB in the filled projectile shall be located radiographically with the projectile oriented in a vertical, nose-up position.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. The filled and closed projectiles shall be packed as shown on Drawing 76-18-44 and segregated by zone before shipment.

5.2 Marking. In addition to any special marking required by the contract or order, palletized shipment shall be marked as specified on Drawing 76-18-44 and in accordance with Standard MIL-STD-129 and the Interstate Commerce Commission regulations.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Filling level. When filled as specified in 3.3, the projectile carries approximately 1.63 pounds of GB. This information is furnished as an aid to procurement and not as a criterion for filling.

6.3 Precautions. Because of the hazards attending the filling of munitions with GB, suppliers are advised to exercise all necessary precautions to protect personnel from injury during filling and handling. Care should be taken to prevent wetting the closure with agent thereby making void the results of the helium test.

6.4 Decontamination. Decontamination can be accomplished by subjecting the projectile to a spray of decontaminating solution (5- to 10-percent commercial grade soda ash, Na_2CO_3) for a minimum period of 1 minute at a minimum pressure of 15 psig in such a manner as to completely wash all surfaces. The projectile should then be rinsed with water to remove all traces of the decontaminating solution. The temperature of the spray and rinse water should be maintained at a range of 140° to 160°F. Spray pressures should have gages with an accuracy of 1 psig. An approved procedure of control of the decontaminating solution strength should be maintained by periodic examination and the time the projectile is subjected to the spray should be automatically controlled and recorded.

6.5 Safety classification. This item is chemical group A (special), explosive hazard class 11 (Chemical Corps Safety Directive No. 385-10).

6.6 Supersession data. This specification supersedes Chemical Corps Purchase Description 196-131-655A, dated 6 March 1956.

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